

Summary Report - Week 19 and Week 20 Activities

During Week 19, the project's focus shifted from model development to advanced evaluation, specifically addressing explainability and cross-dataset generalisation within the VerifAI deepfake detection framework. The primary objective was to assess whether models trained on a specific dataset could generalise to unseen data and to provide qualitative insights into model decision-making processes.

Explainability was addressed by implementing offline Grad-CAM analysis on a frame-level Xception convolutional neural network. Representative frames were extracted from selected videos, and gradient-based activation maps were generated to visualise the spatial regions contributing most strongly to the model's predictions. These visual explanations were saved and documented, demonstrating that the model primarily dealt with facial regions such as the mouth, eyes, and cheeks, rather than irrelevant background areas. This step provided qualitative validation that the model's predictions were based on semantically meaningful visual cues.

Cross-dataset generalisation was evaluated by applying a model trained on the FaceForensics++ (C23) dataset to unseen real-world video data collected independently from controlled datasets. An initial plan to use the KoDF dataset was documented but later abandoned due to storage and hardware constraints. Instead, a real-only external dataset derived from user-recorded survey videos was used to assess model behaviour under domain shift. Predictions, probability scores, and false positive rates were computed and analysed, revealing a high false positive rate on real-only data. This result highlighted the sensitivity of the trained model to distributional differences and reinforced the importance of cross-dataset evaluation when assessing real-world deployment readiness.

In Week 20, the project transitioned to a human-centred evaluation phase, focusing on usability and user trust in the VerifAI system. A controlled user study involving 30 participants was designed and conducted to assess the system's usability using the System Usability Scale (SUS), alongside trust and confidence metrics inspired by prior work on AI trustworthiness. Participants completed a structured interaction task using the VerifAI interface, after which anonymised survey responses were collected.

The collected data included task completion times, error counts, task success indicators, SUS questionnaire responses, and trust-related ratings. These responses were processed using an automated analysis pipeline that computed individual SUS scores and aggregated trust indices. The resulting distributions showed that most participants rated the system as having good usability, with moderate to elevated levels of trust in the model's predictions. Variability in scores reflected realistic differences in user experience and technical familiarity.

The combination of explainability analysis, cross-dataset testing, and user-centred evaluation strengthened the overall validity of the system and demonstrated an awareness of real-world deployment challenges beyond model accuracy alone.